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Understanding organic aerosols in Bogotá, Colombia: In-situ observations and regional-scale modeling

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HIGHLIGHTS

- WRF-Chem simulations and in-situ $PM_{2.5}$ chemical composition data were used to understand the contribution of OA over Bogotá.
- Biomass burning plumes from Northern Amazonia and from the Orinoco River Grassland can severely impact the region.
- Field data shows that OA contributes over 40% to $PM_{2.5}$ in the city.
- The model underpredicts organic aerosols, with the MOZART/MOSAIC scheme predicting higher POA and SOA than RACM/MADE-VBS.

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ABSTRACT

Organic aerosol (OA) is one of the major components of fine atmospheric aerosol particles. Recent field campaigns and modeling studies in Colombia have demonstrated the presence of a large OA fraction in urban and rural aerosols. In this work we focus on constraining the sources associated with OA and its precursors over the city of Bogotá, Colombia. We used $PM_{2.5}$ chemical speciation data from field campaigns carried out during 2018 to evaluate the ability of a regional transport model to reproduce and explain the observed OA. The samples were collected at three sites during high- and low-aerosol concentration seasons in the region. We used The Weather Research and Forecasting Model coupled with Chemistry (WRF-Chem) with two different chemical mechanisms and aerosol schemes, RACM/MADE-VBS and MOZART/MOSAIC, in conjunction with the detailed in-situ aerosol chemical speciation observations to analyze the seasonality of OA in the city of Bogotá, and to establish the most relevant sources. Simulations were carried out for the same periods for which data is available, spanning a high biomass burning (BB) season (February 2018) and a low biomass burning season (September 2018). We demonstrated that long-range transport of BB emissions from Northern Amazonia can episodically increase aerosol loading during September, while BB activity in the Grasslands of the Orinoco River Basin are the main source during February. The two aerosol schemes utilized in this work reproduce the observed seasonal variations in total fine aerosol concentration, with a difference between February and September of 10 $\mu g\ m^{-3}$. The comparison between aerosol schemes demonstrated that MOZART/MOSAIC consistently predicted more OA with a larger SOA:OA ratio than in the RACM/MADE-VBS experiment. SOA dominates the OA fraction by 66% for RACM/MADE-VBS and 74% for MOZART/MOSAIC during February and 69% for RACM/MADE-VBS and 71% for MOZART/MOSAIC during September. These differences between organic aerosol burden between the mechanisms used in this study may be attributed to the different treatment of SOA gas/particle partitioning in the schemes.

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1. Introduction

Organic carbon constitutes the major fraction of atmospheric fine particulate matter in many different environments (e.g., Kanakidou et al., 2005; Ng et al., 2010; Shrivastava et al., 2017), accounting for 20%–60% of the $PM_{2.5}$ in urban areas (e.g., Zhou et al., 2012; Martinez et al., 2012), and for over 90% in tropical forested areas (Kanakidou et al., 2005). Organic aerosol (OA) is a complex mixture of chemical compounds containing carbon–carbon with large variations in composition across locations and time of the year (Forster et al., 2007). Primary organic aerosols (POA) are directly emitted from biogenic and anthropogenic sources such as biomass burning, fossil fuel combustion, or windblown biological particles, among others. Secondary Organic Aerosols (SOA) are mainly formed through the photochemical oxidation of volatile organic compounds (VOCs) via gas-particle transformation (Pandis et al., 1992; Riva et al., 2016). OA has been shown to impact the atmospheric environment, affecting the radiation budget through aerosol-cloud-climate interactions (e.g., Lu et al., 2015; An et al., 2019; Meng et al., 2020; Shrivastava et al., 2017), atmospheric visibility (e.g., Watson, 2002), and human health (e.g., Lelieveld et al., 2015) specially affecting susceptible populations (WHO, 2013).

Despite the number of studies that have demonstrated the relevance of OA for both urban and remote continental aerosols, most of the detailed observations have been carried out in boreal latitudes (e.g., Kanakidou et al., 2005; Jimenez et al., 2009). There are persisting challenges in the understanding of OA sources, its removal processes, and in the chemical formation pathways of SOA. The latter is a major fraction of global submicron-sized atmospheric OA (Shrivastava et al., 2017). To overcome these limitations, research groups around the world have studied OA abundance and formation mechanisms using detailed observational campaigns (e.g. Hayes et al., 2013; Mancilla et al., 2015; Rastogi et al., 2014) and laboratory experiments (e.g. Surratt et al., 2006). Some of these experimentally constrained parameters have been incorporated into chemical transport models, which are used to evaluate the potential impact that these formation mechanisms have on the observed ambient concentration of OA. However, little is known about the prevalence of OA in tropical areas, and the relative importance of SOA and POA are yet undetermined in many urban areas in these regions. Continental equatorial areas can have substantial emissions of biogenic VOC from vast forested areas and agricultural land, and, in the larger cities of the region, anthropogenic emissions can also be a significant source of POA and SOA precursors. Recent work has shown a significant regional impact of biomass burning (BB) plumes in the northern part of South America, especially during the dry season of January–April, when BB activity is high. These BB plumes, mainly coming from the grasslands of the Orinoco River Basin, can affect the densely populated cities in the region which are often hundreds of kilometers away from the BB source regions (Ballesteros-González et al., 2020; Mendez-Espinosa et al., 2019; Hernandez et al., 2019). These BB events in northern South America have different temporal patterns to those in Amazonia. The latter have been studied more extensively. This seasonality in BB partly explains the strong observed $PM_{2.5}$ annual cycle in the region (Ballesteros-González et al., 2020; Hernandez et al., 2019) together with the prevailing seasonal meteorological conditions (Mendez-Espinosa et al., 2019). Although millions of people can potentially be impacted by BB smoke, only a handful of studies have collected detailed aerosol chemical speciation data or used regional-scale modeling in this region (e.g. Rincón-Riveros et al., 2020). Observational campaigns in Bogotá, Colombia show a persistent OA fraction over the city (e.g. Vargas et al., 2012; Ramírez et al., 2018; Rincón-Riveros et al., 2020). Vargas et al. (2012) suggested that OA was mainly of secondary origin based on EC to OC ratios. Recently, Rincón-Riveros et al. (2020) showed a strong association between OC and the BB tracer levoglucosan, during the dry season, and a strong correlation of Brown Carbon with the number of fires in the region. Ramírez et al. (2018) characterized PM_{10} chemical composition during 1 year at an urban background site in Bogotá. The authors found that carbonaceous material constitutes 51% of PM_{10} at the site, with a large OA fraction persistent throughout the year, i.e., not only during the high BB period of January to April. This suggests there might be other unaccounted sources of OA and its precursors in this region.

To better constrain the sources associated with OA and understand its temporal and spatial variation in the Northern part of South America, we used a chemical transport model (WRF-Chem), which considers the secondary generation and the interactions between several species in the atmosphere (e.g., Grell et al., 2005), in conjunction with detailed in-situ observations of chemical $PM_{2.5}$ composition over Bogotá. In this study we use modal and sectional aerosol schemes with different treatments of SOA formation to analyze their ability to reproduce observed OA and biomass burning tracers analyzed during the field campaigns. The observational data was collected during periods of typically high and low atmospheric aerosol concentration in the region.

2. Methods

2.1. Area of study

Bogotá, the capital of Colombia, is located in the equatorial Andean region of South America. The city, with a population of 8 million, sits in a plateau 2600 m above sea level. This plateau is located to the east of the Magdalena River valley, and is surrounded on its eastern and southern margins by mountains reaching up to 4000 m (Fig. 1a). Its relatively small urban area, of 380 km^2, makes Bogotá one of the most densely populated cities in Latin America (Wheeler, 2015). Observations from the local Air Quality Monitoring Network (AQMN) show a $PM_{2.5}$ annual mean of 18.2 $\mu g\ m^{-3}$ in 2018, which exceeds the World Health Organization (WHO) guidelines (WHO, 2006). However, the city experiences a strong seasonal variation in $PM_{2.5}$, causing much higher daily $PM_{2.5}$ values from January to March (Mendez-Espinosa et al., 2019), often triggering air pollution alarm episodes during those months. The 24-h $PM_{2.5}$ exceeded the daily local limit of 37 $\mu g\ m^{-3}$ 125 times during 2018 and 256 during 2019. Most of those exceedances occur in the south-west of the city were industrial sources and heavy duty traffic are prevalent (East et al., 2021). Concentrations are much lower from July to September. In this study we use aerosol composition data collected at three sampling sites in the city during low- and high- BB activity seasons. The chemical transport model WRF-Chem was also configured to run simulations on this region, for simulation periods matching those of the field campaigns. The modeling domain covers the northern half of South America, including relevant BB source areas like the Orinoco River Grasslands, and Northern Amazonia (Fig. 1).

2.2. $PM_{2.5}$ chemical speciation data

In this work we use $PM_{2.5}$ chemical speciation data obtained during field campaigns carried out in the city of Bogotá during 2018 and 2019. Two sampling sites, *Hacienda Noviciado* - **HN** (lat. 4,836; lon. −74,092) and *Monserrate* - **MT** (lat.4,605; lon.-74,055), were deployed at suburban areas. MT is located 550 m higher than the average elevation of the city. These locations, which are outside the urban perimeter, were selected to detect possible influences from regional atmospheric transport (Fig. 1a). A third sampling site *Universidad Libre* - **UL** (lat. 4,666; lon. −74,102) was located near the urban geometric center to characterize urban background concentrations. The UL site is 500 m away from major roads and no industrial activities are located in the area (Fig. 1a). Sampling campaigns were carried out during low and high pollutant concentration seasons of 2018 according to the known air quality annual cycle in the region. The first campaign (C1) was carried out between January 25 and April 07 at the HN and MT sites, collecting 59 24-h $PM_{2.5}$ samples during the high pollutant concentration season. The second campaign (C2) was carried out between July 25 and September 26 at HN, UL, and MT during a low pollutant concentration season, collecting

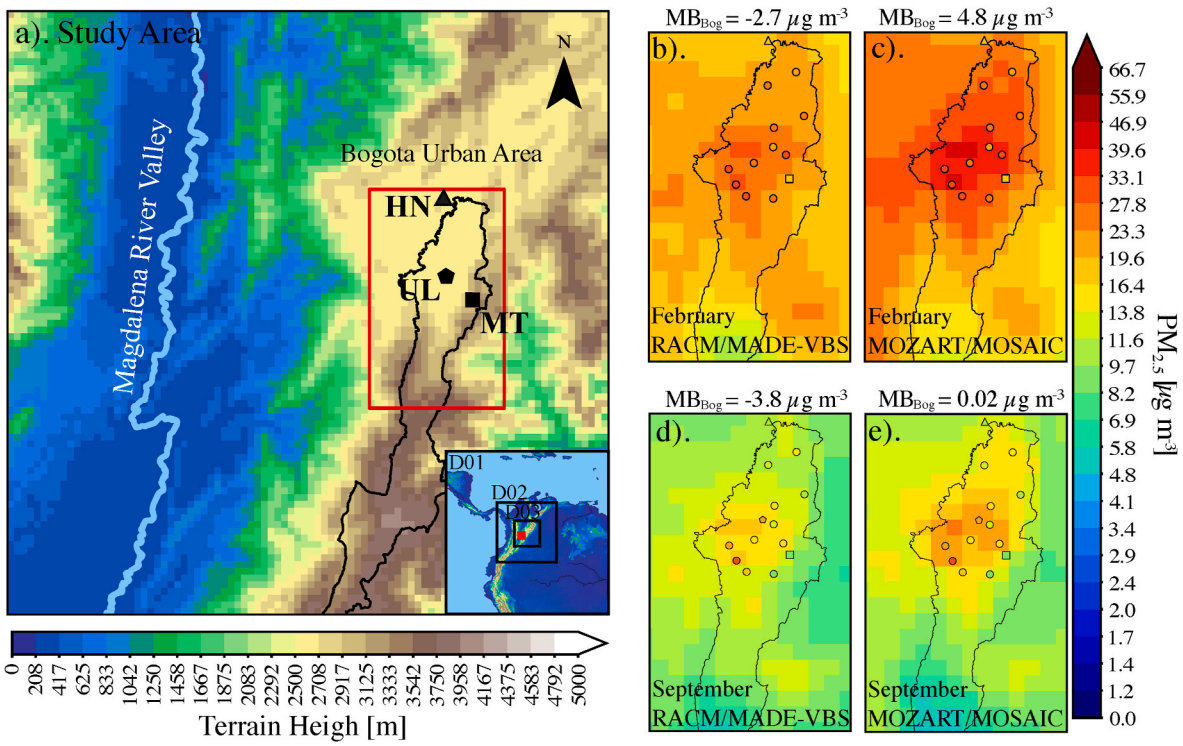


Fig. 1. a). Terrain height in the area of study and location of the sampling sites within Bogotá city limits. HN: Hacienda Noviciado, UL: Universidad Libre, and MT: Monserrate sampling site. The inset depicts the modeling domains used in this study (D01, D02, and D03). Monthly mean ground level $PM_{2.5}$ concentrations from modeling and observations are shown for February 2018 and for both b). RACM/MADE-VBS, c). MOZART/MOSAIC, and during September 2018 d). RACM/MADE-VBS and e). MOZART/MOSAIC. Monthly Mean Bias (MB) are presented at the top of each concentration map experiment.

a total of 76 samples. These periods also correspond to high (January–April) BB emissions in Northern South America and low BB emissions (June–September) (Fig. 6). Sample collection and analysis is described in detail in Rincón-Riveros et al. (2020), and will be briefly summarized here.

Samples were collected on quartz filters using low- and high-volume samplers, with flow rates of 16.6 and 1140 LPM respectively. Before its field deployment, filters were placed in a muffle at 550 °C for 12 h to reduce the organic background (Bao et al., 2019; Graham et al., 2002). After collecting the sample, filters were stored in a desiccator and then weighted under controlled temperature and humidity. Samples were analyzed for organic carbon (OC), elemental carbon (EC), water-soluble organic carbon (WSOC), carbohydrates, and inorganic ions at the Collett Laboratory at Colorado State University. OC and EC were determined with a thermal–optical transmission EC/OC analyzer (Sunset Labs Inc.). A portion of the filter was extracted in deionized water. The WSOC of the extract was quantified using a total organic carbon (TOC) analyzer (Sievers model M9 Turbo) (Sullivan et al., 2011). An aliquot of the aqueous extract was also used to analyze for levoglucosan, galactosan, mannosan and other anhydro-sugars using high-performance anion-exchange chromatography with pulsed amperometric detection (HPAEC-PAD) (Sullivan et al., 2011). Inorganic ions such as Sulfate (SO_4^{2-}), Nitrate (NO_3^-), Ammonium (NH_4^+), Potassium (K^+), among others, were analyzed from another aliquot of the aqueous extract. Cations and anions were analyzed using appropriate ion-exchange columns. Complete method details are presented by Sullivan et al. (2008, 2011).

Hourly surface measurements of criteria pollutants from the air quality monitoring network of Bogotá (AQMN) were also used during the field campaign periods.

2.3. WRF-chem model configuration

In the WRF-Chem model, atmospheric chemistry processes are coupled to the meteorological fields of the regional scale numerical weather prediction model WRF (Grell et al., 2005; Fast et al., 2006; Powers et al., 2017). Model configuration was selected based on the skill shown on previous simulation studies of regional transport of biomass burning plumes over the region (Ballesteros-González et al., 2020). The full details of each physics option and configuration are summarized in Table 1.

To investigate the model sensitivity to the aerosol scheme used we set-up two experiments, RACM/MADE-VBS (termed EXP1) which uses a modal aerosol scheme, and MOZART/MOSAIC (EXP2 hereafter) which uses a sectional approach for aerosol particles. For these numerical ex-

Table 1
WRF-Chem model parameterization.

Parameter	Scheme		Reference
	EXP1	EXP2	
Microphysics	Lin et al.	Lin et al.	Lin et al. (1983)
Longwave Radiation	RRTM	RRTM	Mlawer et al. (1997)
Shortwave Radiation	Dudhia	Dudhia	Dudhia (1989)
Land Surface	Noah	Noah	Ek et al. (2003)
Planetary Boundary Layer	YSU	YSU	Hong et al. (2006)
Cumulus	Grell-Freitas	Grell-Freitas	Grell et al. (2014)
Gas-phase Chemistry	RACM[♯]	MOZART[♯♯]	[♯]Stockwell et al. (1997) [♯♯]Emmons et al. (2010)
Aerosol Mechanism	MADE[♭]	MOSAIC[♭♭]	[♭]Ackermann et al. (1998) [♭♭]Zaveri et al. (2008)
SOA Scheme	VBS[♮]	Simple Scheme [♮♮]	[♮]Ahmadov et al. (2012) [♮♮]Hodzic and Jimenez (2011)

periments we selected aerosol schemes capable of simulating the chemical composition across the aerosol size distribution. EXP1 was carried out with MADE-VBS which is the Modal Aerosol Dynamics model for Europe (Ackermann et al., 1998) with a SOA formation mechanism based on the Volatility Basis Set approach (VBS) (Ahmadov et al., 2012) and the RACM gas-phase chemistry module (Stockwell et al., 1997Stockwell et al., 1997). The MADE aerosol scheme describes the aerosol size distribution with 3 log-normal aerosol modes: the Aitken mode (<0.1 μm diameter), accumulation mode (0.1–2 μm diameter), and coarse mode (>2 μm diameter) (Ackermann et al., 1998). SOA formation in MADE is described with the VBS, which categorizes organics species into volatility bins and uses partitioning theory to estimate gas-phase and particle-phase amounts (Donahue et al., 2009Donahue et al., 2009). RACM, a lumped-molecule gas-phase chemistry mechanism, is an update of the second generation Regional Acid Deposition Model and includes 78 chemical species and 214 gas-phase reactions (Stockwell et al., 1990, 1997).

The EXP2 uses the Model for Simulating Aerosol Interaction and Chemistry (MOSAIC) aerosol model (Zaveri et al., 2008) and the MOZART gas-phase mechanism (Emmons et al., 2010). MOSAIC uses a sectional approach dividing aerosol particles into four internally-mixed discrete size bins from 0.039 μm to 10 μm (Zaveri et al., 2008). In this chemical option, SOA is estimated by considering an organic gas as a lumped SOA precursor which is allowed to react with hydroxyl radicals (OH) into a single non-volatile species that condenses to form SOA. The SOA condensation step uses a parameterization based on the observed proportionality of SOA concentrations to excess carbon monoxide (CO) and photochemical age of the air mass (Hodzic and Jimenez, 2011). The MOZART gas-phase chemistry mechanism includes 196 chemical reactions and 157 gas-phase reactions (Emmons et al., 2010). The MOSAIC aerosol scheme considers the major aerosol components including sulfate, nitrate, ammonium, sodium, black carbon, and primary organics, among other species. Other unspecified inorganic species are lumped as other inorganic mass (OIN), such as silica, minerals and trace metals (Zaveri et al., 2008).

Simulations were conducted for the months of February and September of 2018 for both chemical mechanisms with 48-h spin-up time. These simulation periods were selected to match periods when BB activity peaks in the region (February) and a period of near minimum BB in Northen South America (September) (Mendez-Espinosa et al., 2019) (Fig. 6). These simulations periods also match the field campaigns carried out to collect detailed $PM_{2.5}$ chemical speciation data at various locations in Bogotá (Fig. 1).

Three nested modeling domains are used to cover the northern half of South America to dynamically downscale boundary conditions from global transport models. The domains centered in Colombia (lat. 5.194 – long. −73.522) have a horizontal resolution of 27x27, 9x9, and 3 × 3 km for domain 1 (D01), domain 2 (D02), and domain 3 (D03), respectively. The city of Bogotá is within the higher resolution domain DO3. The 3 km horizontal resolution for D03 is necessary to capture the complex topography near the city. We used forty-one vertical levels in the simulations. To ensure sufficient vertical resolution to capture the boundary layer dynamics, fourteen vertical levels were allocated to the 1 km layer closest to the surface. The urban canopy parameterization available in WRF-Chem was not considered in this study as the spatial resolution of D03 was considered too coarse to resolve urban canopy effects (Salamanca et al., 2011; Mauree et al., 2018; Ribeiro et al., 2021). National Center for Environmental Prediction Final Operation Global Analysis (NCEP-FNL) provided the initial and boundary conditions for the meteorological fields. NCEP-FNL has a resolution of 1° and 6-hourly temporal resolution (NCEP, 2000). Initial and boundary conditions for chemical species were extracted from the global chemical transport model CAM-Chem (Buchholz et al.), and mapped to the WRF-Chem outermost domain boundaries. CAM-Chem data has a horizontal resolution of 0.9° × 1.25° and 56 vertical levels. Boundary and initial conditions for the inner domains were passed down from the respective parent domains. Numerical experiments shared the same model configuration and physical parameterizations.

2.4. Emissions

Anthropogenic, biogenic, and biomass burning (BB) emissions are included in the simulations. Anthropogenic emissions from the Emissions Database for Global Atmospheric Research version 4.3.1 (EDGAR V4.3.1) were used (Crippa et al., 2016). EDGAR V4.3.1 has a horizontal resolution of 0.1° × 0.1°. The anthropogenic emissions from EDGAR were assumed constant for both simulation periods. Activity patterns in the tropics do not exhibit the large variations typical of higher latitudes. For the innermost domain, we combined emissions from a local emission inventory for Bogotá city limits (Pachón et al., 2018) with those of EDGAR for the rest of the domain. The Bogotá local emission inventory includes emissions from commercial, mobile, and industrial sources. For PM, also re-suspended particulate matter (RPM) sources are considered. The spatial resolution for the local emission inventory is 1 km × 1 km. RPM emissions were modified for each modeling period according to a mitigation factor developed by Pérez-Peña et al. (2017). Additionally, we tuned-down gas-phase emissions from mobile sources as those are thought to be overestimated by the local emission inventory (Gallardo et al., 2012). We used 35% of nitrogen oxides ($NO_x = NO + NO_2$) and VOCs, and 45% of CO from this emission source. Different speciation profiles were applied to each type of source (supplementary material S1). Mobile source emissions were separated according to fuel use share in Bogotá, where 54% is diesel and 46% is gasoline. Separate diurnal activity profiles were applied to the emissions from commercial, industrial, mobile and RPM sources respectively. Anthropogenic emissions were injected at ground level, as stack height information is not available in the local emission inventory. However, point sources are relatively minor within the city for reactive gases and primary particles. Biomass burning emissions were taken from the Fire Inventory from NCAR, FINN version 1 (Fig. 6) (Wiedinmyer et al., 2011). FINN has hourly temporal resolution and spatial resolution of 1 km × 1 km. Previous studies in the same domain suggests FINN emissions are more accurate than those of other emission inventories (Ballesteros-González et al., 2020). The plumerise module available in WRF-Chem was utilized for fire emissions. Biogenic emissions are calculated online using the Model of Emissions of Gases and Aerosols from Nature, MEGAN (Guenther et al., 2012), with emissions rates varying according to meteorological variables.

3. Results

Modeled meteorological fields were compared against AQMN surface observations in Bogotá. A detailed model performance evaluation is presented in supplementary material S2. Meteorological parameters for RACM/MADE-VBS and MOZART/MOSAIC experiments perform similarly well, with only minimal differences among configurations. Other studies have also reported negligible, but not zero, differences between two simulations without aerosol-radiation interactions (e.g., Mar et al., 2016; Karlický et al., 2017). Temperature mean gross error (MGE) and index of agreement (IOA) meet the criteria values recommended by Emery et al. (2001) for all simulations. The modeling results underestimate temperature by 1.8 K during February and by 1.1 K during September. The underestimation is larger during daytime (supplementary material S3). Wind speed and direction are well simulated during both simulation periods (Fig. 2g, 2h, 2i, and 2j). In both model configurations, downward shortwave (SW) radiation at the surface is overestimated (supplementary material S10), suggesting that cloud cover is not well captured in the simulations. Cloud cover underestimation can enhance both, the downward SW flux (impacting photo-chemistry) and the radiational cooling, and could help explain the modeled temperature bias.

Simulated wind speed meets the goal benchmark for RMSE and IOA

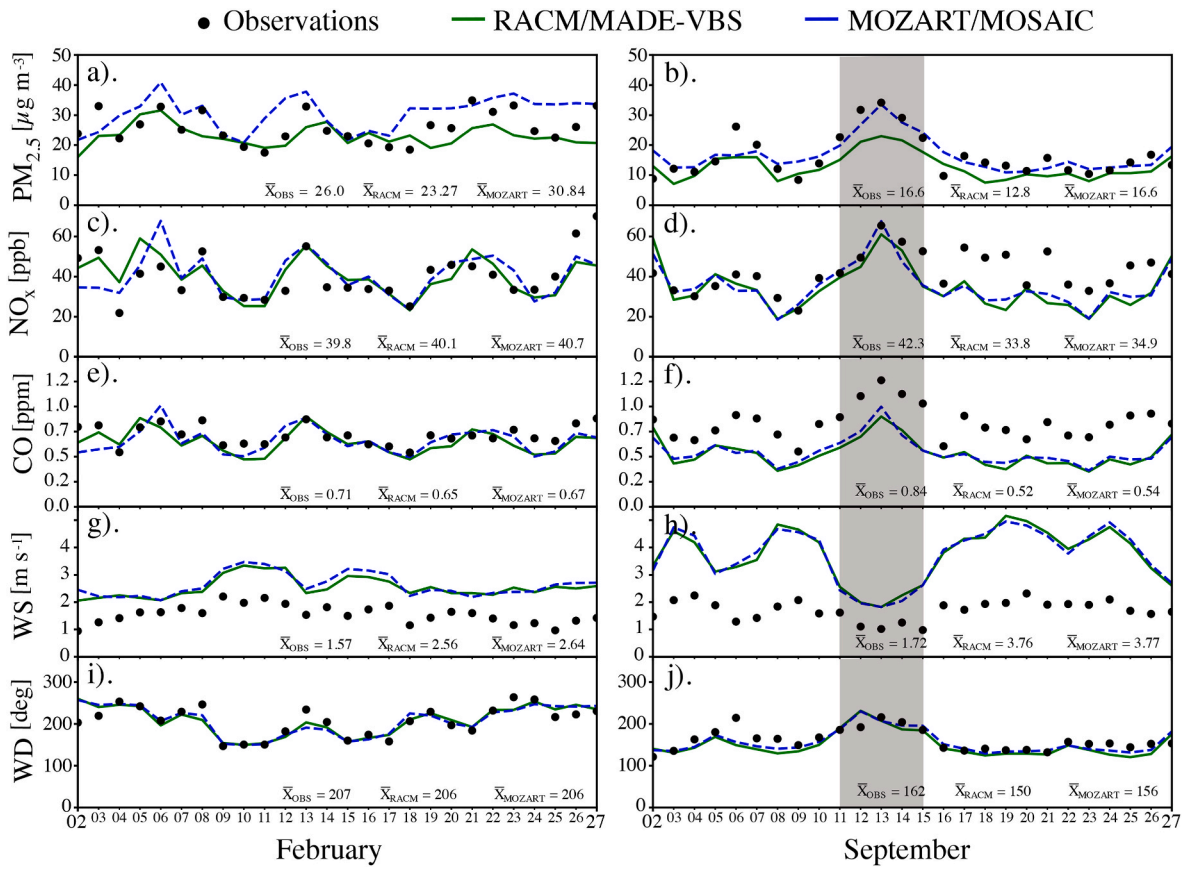


Fig. 2. Time series comparison of air pollutants measured in Bogotá and simulation results RACM/MADE-VBS (green line) - MOZART/MOSAIC (blue line) during February 2018 (left panels) and September 2018 (right panels); (a, b) $PM_{2.5}$, (c, d) Nitrogen Oxide, (e, f) carbon monoxide, (g, h) wind speed, and (i, j) wind direction. Gray shaded region highlight period with high $PM_{2.5}$, NO_x, and CO concentrations during the low BB activity season. This event is explained in discussion section. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

during February and meets all the benchmark criteria during September. Observed monthly mean wind speed during September is 15% higher than February. The model wind field is biased high by 1 m s^{-1}. However, due to the low wind speeds in the city, this bias is equivalent to a 45% relative bias. These higher wind speeds imply the model is more efficient in transporting pollutants, thereby inducing an underestimation in pollutant concentrations.

3.1. $PM_{2.5}$ model performance

Simulations exhibit a seasonality similar to that of observations. The difference between the observed monthly mean $PM_{2.5}$ concentration for February and September is 9.4 μg m^{-3}. Both modeling experiments show similar seasonality, with a February to September difference of 10.5 μg m^{-3} and 14.2 μg m^{-3} for RACM/MADE-VBS and MOZART/MOSAIC, respectively (Fig. 2a and b). Previous studies suggest the elevated $PM_{2.5}$ levels in February might be explained by regional transport of BB plumes (e.g., Rincón-Riveros et al., 2020; Ballesteros-González et al., 2020). September simulations show a 1 m s^{-1} increase in wind speed, which partially explains the lower concentrations, together with a lower regional background from decreased BB emissions during this period.

The model monthly-mean spatial distribution of $PM_{2.5}$ correctly reproduces the locations with high concentrations within the city, which are zones dominated by industrial sources and heavy duty traffic (Fig. 1). Modeled $PM_{2.5}$ concentrations in the surroundings of the city are around 33% and 43% lower than over urban areas for February and September, respectively, highlighting the impact from anthropogenic sources at the local scale.

We compared the observed and modeled $PM_{2.5}$ chemical composition at the three sampling sites (Fig. 3). C1, carried out during the high-BB activity season, shows higher $PM_{2.5}$ than those over the low-BB activity season. The observed concentration at the suburban sampling sites (MT and HN) is a factor of two lower than at the UL urban site. Furthermore, the observed OA fraction was high across all sampling sites, ranging from 38% at HN to 65% at the MT. Similarly large organic fractions were reported in Mexico city (Martinez et al., 2012).

The MT site reported the highest OA fraction in both campaigns. At the HN site during C1 unspecieted $PM_{2.5}$ had the highest contribution to $PM_{2.5}$ with 45% of the total, followed by OA with 35% of the total. Although biomass burning activity was much lower during C2, the chemical composition had a similar distribution as that of C1. Additionally, at the HN site, a significant contribution of SO_4^{-2} was detected during both campaigns. At the urban site UL, OA was also the dominant fraction with 48%, followed by EC with 23% and a lower presence of unspecieted $PM_{2.5}$ (OTH). The large contribution of EC at this site suggests it is highly impacted by traffic related emissions (Fig. 3).

Consistently with field campaign data, simulations show that the major aerosol contributors to $PM_{2.5}$ in the city are OTH and OA (Fig. 3). Resuspended dust is known to be an important source of $PM_{2.5}$ in the city (e.g., East et al., 2021; Pachón et al., 2018) and in the model their are included in OTH component. However, the OA fraction is underestimated in most cases on average by 20%. Only the MOZART/MOSAIC experiment at the HN location overestimates the OA fraction during February. While EC is well captured by the model at the HN and MT sites, both modeling experiments underestimate this component by a factor of three, equivalent to missing 3.3 μg m^{-3}, at the UL site. The underestimation of EC at this urban site likely points to missing EC emissions in the local inventory. Small discrepancies for sulfate and

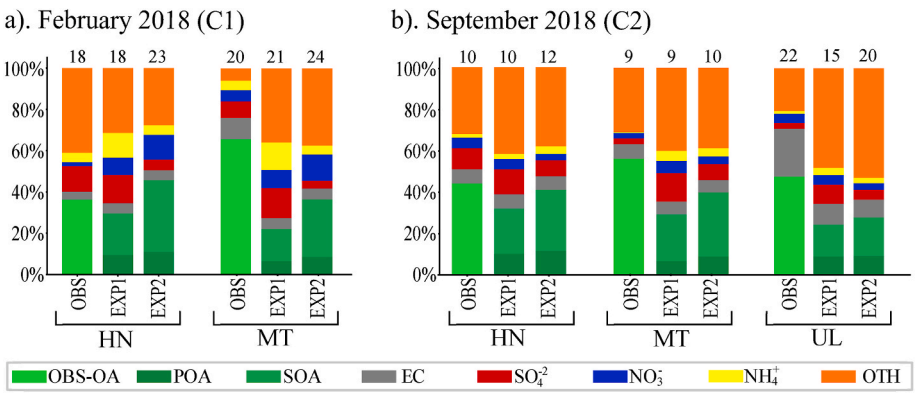


Fig. 3. Field campaign $PM_{2.5}$ chemical speciation results for the month of February (C1) and September (C2) against simulation results at three sampling site locations. Total $PM_{2.5}$ mass at each location is shown at the top of each bar in $\mu g\ m^{-3}$. OA mass was estimated based on Turpin and Lim (2001) factors, we used 1.6 ± 0.2 for urban site UL and 2.1 ± 0.2 for rural sites HN and MT.

nitrate were observed between aerosol schemes. Considering that the two experiments share the same emission inventories and physical parameterizations, differences should be caused by variations in aerosol dynamics and in the allocation of aerosol mass between log-normal modes and bins.

The model captured $PM_{2.5}$ peak concentrations (Fig. 2a and b), accurately representing the evolution of fine aerosol mass concentrations for the two seasons. Fig. 4 compares the daily mean $PM_{2.5}$ chemical speciation with in-situ observations showing that during February 6, 13 and 18 at MT location (Fig. 4c) and September 6 and 13 at UL location (Fig. 4a) there is an increase on $PM_{2.5}$ that is dominated by the OA fraction. These peaks are better represented by the MOZART/MOSAIC experiment at both locations. Simulation results highlight that the SOA is the component with the highest increase during these peaks with 27% and 22% for February and September peaks, respectively (Table 2).

Qualitatively consistent with available observations, OA was prevalent across all simulations (supplementary material S4). During February, the modeled OA fraction is 24% and 35% and during September 25% and 31% for RACM/MADE-VBS and MOZART/MOSAIC, respectively. The POA fraction remains constant during both

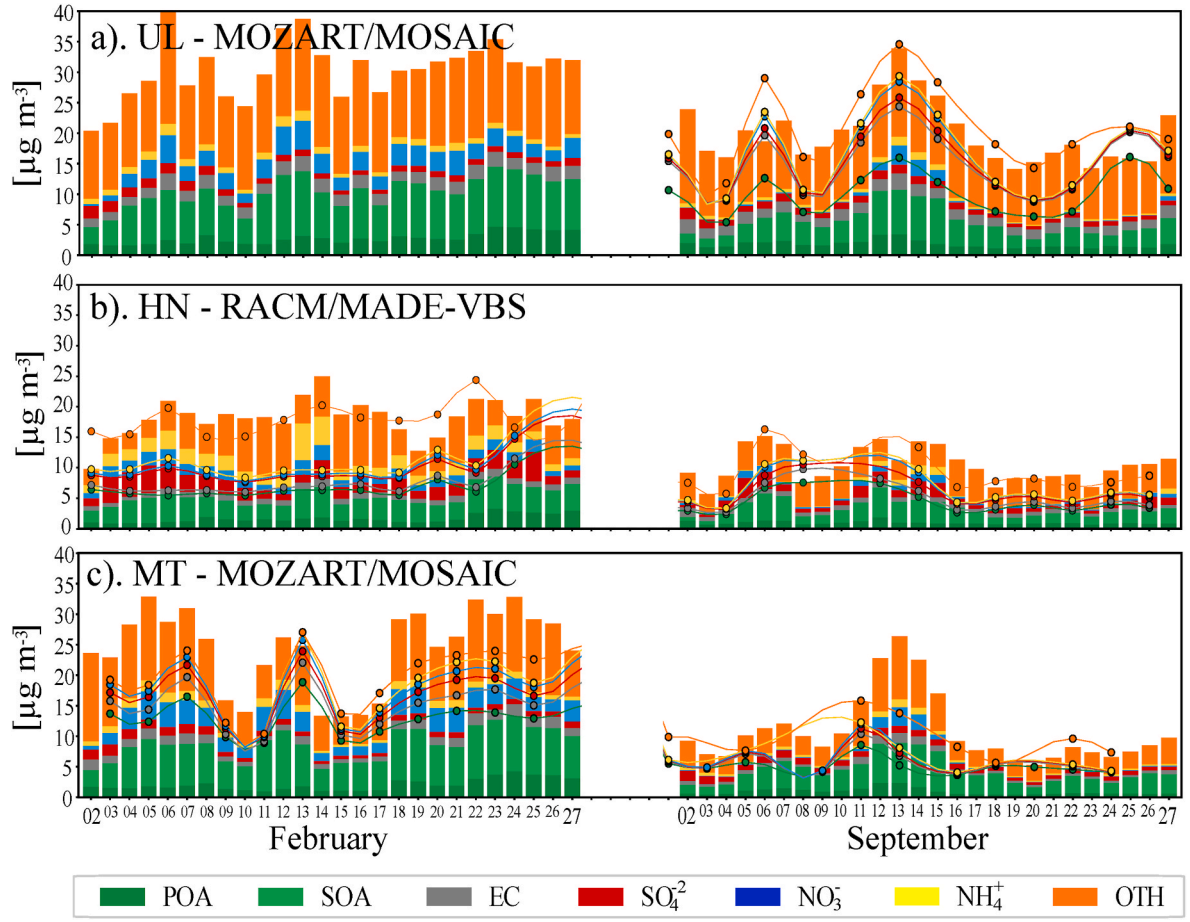


Fig. 4. Daily mean $PM_{2.5}$ chemical speciation for the month of February and September 2018. Bars represent model simulation results and lines represent field campaigns results.

Table 2
Monthly mean aerosol concentrations from observations at HN, MT and UL sites against modeling results for RACM/MADE-VBS and MOZART/MOSAIC chemical parameterization. Values marked with * are observed OA mesured during field campaings, which cannot be directly decomposed into POA and SOA.

		Observ.		EXP1		EXP2	
Sampling Site	Aerosol-species	Feb	Sep	Feb	Sep	Feb	Sep
HN	POA	*6.51**	*4.38**	1.70	1.01	2.55	1.39
	SOA			3.62	2.26	7.97	3.63
	EC	0.68	0.69	0.9	0.73	1.12	0.79
	SO_4^{-2}	2.22	1.01	2.47	1.23	1.18	0.96
	NO_3^-	0.37	0.51	1.51	0.52	2.76	0.37
	NH_4^+	0.80	0.17	2.14	0.27	1.06	0.45
	OTH	7.35	3.23	5.67	4.33	6.39	4.71
MT	POA	*12.83**	*5.08**	1.43	0.59	2.09	0.95
	SOA			3.18	2.08	6.60	3.36
	EC	2.04	0.64	1.10	0.58	1.26	0.64
	SO_4^{-2}	1.54	0.26	2.99	1.25	0.86	0.85
	NO_3^-	1.07	0.24	1.8	0.55	3.03	0.41
	NH_4^+	0.87	0.04	2.74	0.45	1.01	0.42
	OTH	1.14	2.81	7.37	3.68	8.79	4.21
UL	POA		*10.28**		1.31		1.81
	SOA				2.24		3.67
	EC		4.98		1.46		1.69
	SO_4^{-2}		0.59		1.34		0.94
	NO_3^-		0.98		0.66		0.63
	NH_4^+		0.27		0.53		0.52
	OTH		4.45		6.99		10.42

simulations periods, 8% for RACM/MADE-VBS and 9% for MOZART/MOSAIC. Simulations suggest that SOA is the dominant component in OA, with 66% for RACM/MADE-VBS and 74% for MOZART/MOSAIC during February and 69% for RACM/MADE-VBS and 71% for MOZART/MOSAIC during September. MOZART/MOSAIC consistently simulates more SOA than RACM/MADE-VBS. This differences between SOA within mechanisms may be attributed to MOZART/MOSAIC containing a simple SOA gas/particle partitioning scheme and has more biogenic specifies included.

SO_4^{-2}, NO_3^-, and NH_4^+ ground level concentrations over Bogotá for both simulations are presented in supplementary material S6. The monthly mean SO_4^{-2} concentration was generally lower in the MOZART/MOSAIC experiment (2.29 μg m^{-3} for February and 0.82 μg m^{-3} for September) than in the RACM/MADE-VBS experiment (3.52 μg m^{-3} for February and 1.85 μg m^{-3} for September). These differences may be due to the SO_2 oxidation pathways in each scheme. RACM considers the aqueous-phase oxidation of SO_2 by hydrogen peroxide (H_2O_2) while MOZART does not (Aan de Brugh et al., 2011). NO_3^- concentration in both experiments shows similar performance with higher concentration in MOZART during February (MOZART/MOSAIC = 3.38 μg m^{-3} and RACM/MADE-VBS = 2.08 μg m^{-3}) and higher concentrations in RACM/MADE-VBS during September (MOZART/MOSAIC = 0.47 μg m^{-3} and RACM/MADE-VBS = 0.73 μg m^{-3}). NO_3^- concentration is mainly determined by its precursor HNO_3 (Yang et al., 2018Yang et al., 2018). MOZART presents higher nitric acid (HNO_3) concentrations over Bogotá in both simulations periods, this is due to the MADE aerosol scheme adopting the Mozurkewich (1993) approach, which appears to be more effective than the Zaveri et al. (2008) approach in the MOSAIC aerosol scheme (e.g Yang et al., 2018; Balzarini et al., 2015). For the NH_4^+ monthly mean concentrations during February RACM exhibited higher values than MOZART with 2.81 μg m^{-3} and 1.22 μg m^{-3}, respectively. During September NH_4^+ concentrations are lower than February but the MOZART/MOSAIC experiments led to higher values than RACM/MADE-VBS with 0.51 μg m^{-3} and 0.38 μg m^{-3}, respectively.

4. Discussion

When the temporal evolution of the surface $PM_{2.5}$ is compared to observations, the day-to-day variations during September are better captured than those in February as shown by the correlation of these fields (Fig. 2). The modeled monthly-mean $PM_{2.5}$ for September is closer to observations than those of February.

The agreement between modeled and observed day-to-day variations during September can be explained by the success of the simulations in capturing two periods with high $PM_{2.5}$, NO_x, and CO concentration. These periods occurred between September 4 to 7, and then again from 11 to 15 of September (Fig. 2b, 2d, and 2f). Both, the model results and observational data show a sustained build up of pollutant concentrations in the city. $PM_{2.5}$ increased from 8 μg m^{-3} to 34 μg m^{-3} between September 9 to 13, with the concentrations decreasing rapidly after that episode. Analysis of modeled wind fields over this period show a wind direction change, from easterly to westerly winds over the city, together with lower wind speeds during this period (Figs. 2 and 5). The wind direction change and the decrease in wind speed suggest some of this high-pollution event might be due to a stagnation period. In the periods before and after the September 11–15 event, easterly winds were prevalent over Bogotá, and increases were observed. However, the simulations show there is a regional increase in aerosol loading in all modeling domains during this period, indicating that this event is not simply the result of a local atmospheric stagnation phenomenon. This is clear from the difference in the modeled SOA concentrations during these periods (Fig. 5).

Field data show that the increased $PM_{2.5}$ during September 4 to 7, and September 11 to 15 is dominated by organic aerosols (Fig. 4). The Modeled SOA:OA ratio is higher during such periods. Regional-scale wind fields show southerly winds prevailing over the study area, advecting SOA and CO from the south of the domain (Fig. 5). An analysis of biomass burning emissions shows an increase in intensity of activity in Northern Amazonia, at the south of the domain during September (Fig. 6). This suggests that the transport of BB plumes northwards from Amazonia through the Magdalena River Basin can impact the aerosol concentrations in the cities in the study region, especially SOA, even during periods of low BB activity in the grasslands of Northern South America (Fig. 5). The long-range transport of pollutants is further confirmed by analyzing transport outside our outermost domain, D01. The Global chemical transport model CAM-Chem shows POA, SOA and mineral dust being advected through the eastern boundary of the domain (supplementary material S5). The POA and SOA are likely originating from BB emission areas further south in the Amazon, and even from western-equatorial Africa, suggesting that long-range transport is a significant source of OA during these periods. Our results indicate that although BB emissions in the modeling domain are almost an order of magnitude higher during February than September, the intense sources in Northern Amazonia or even further to the south, do indeed have the potential to impact air quality in Northern South America. Therefore, in addition to the local sources of POA and SOA precursors, long-range transport of Amazonian BB plumes even in the low BB activity season, together with trans-Atlantic transport might contribute to OA in the city, and are consistent with the observed prevalence of BB tracers over this period. Other observational studies in the region had previously demonstrated the potential for long-range transport of aged BB plumes from Amazonia and even the possibility of trans-Atlantic transport of BB from Africa towards the northern part of South America Hamburger et al. (2013).

Consistent, with modeling results and with the proximity and intensity of BB sources during February, levoglucosan concentrations were higher during C1 than C2. Mean levoglucosan concentrations were 77 ng m^{-3} at the HN and 33 ng m^{-3} at the MT sites during February. During September, they were lower, with 6 ng m^{-3}, 44 ng m^{-3}, 5 ng m^{-3} for HN, MT, and UL, respectively. During the September 11–17 period, levoglucosan concentration increased across the three sampling locations.

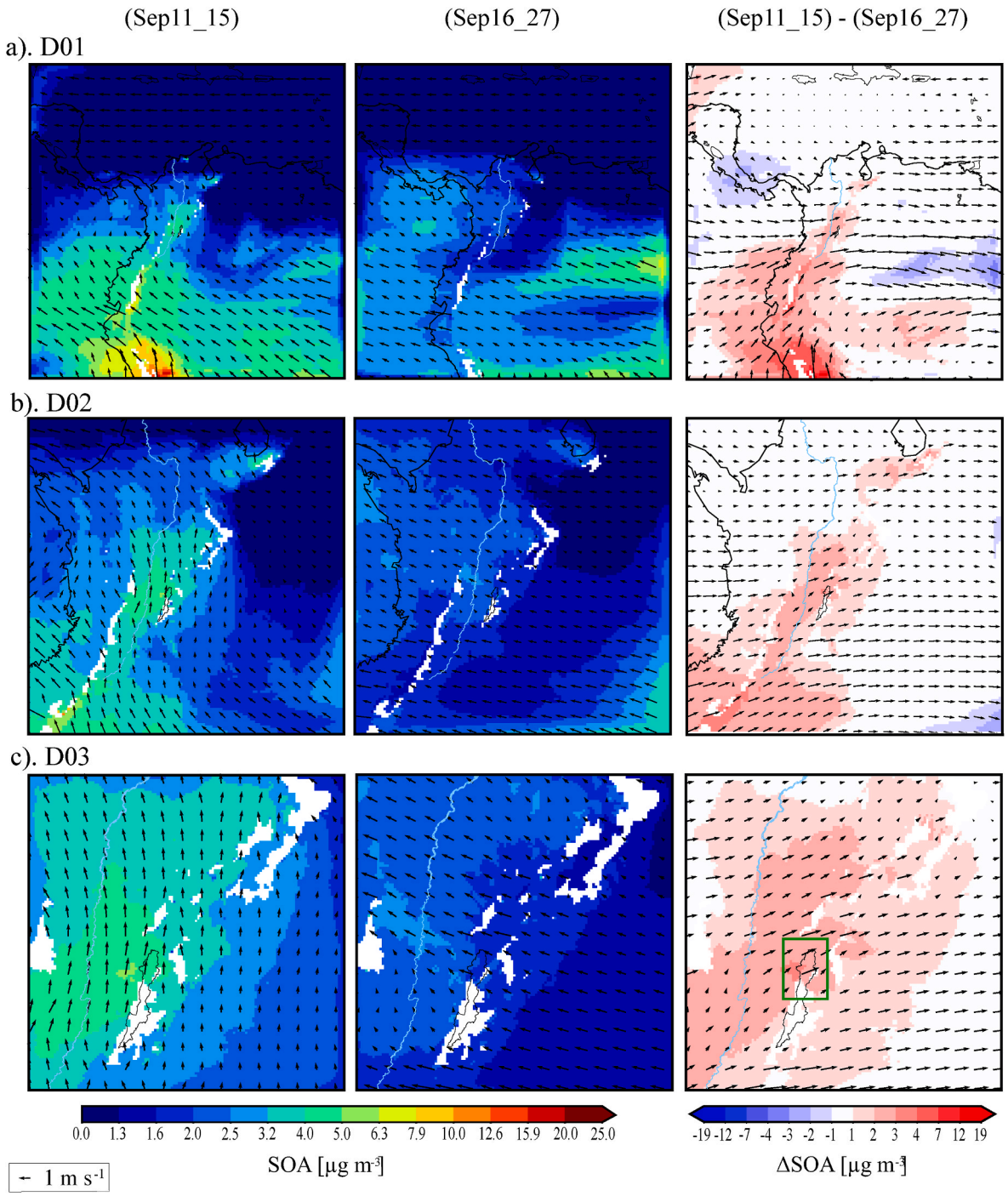


Fig. 5. SOA concentrations from the RACM/MADE-VBS experiment and wind interpolated at 700 hPa from September 11 to 15 (1st column) and September 16 to 27 (2nd column), and difference in these two periods (3rd column) for the three modeling domains.

The mean concentration went from 26 ng m^{-3} before this period to 50 ng m^{-3} during the September 11–17 episode, reaching concentrations similar to those reported during high-BB season (supplementary material S7).

Regarding other potential missing sources of carbonaceous aerosol in the city, the three-fold underestimation of EC at the UL urban site points to missing primary emissions, both from EC and OC, in the local inventory. This suggests that better constrained emissions and speciation profiles should be considered in further research. Furthermore, recent experimental work has shown that the PM_{10} fraction of resuspended dust contains both EC and OC (Ramírez et al., 2019). In the study developed by Ramírez et al. (2019) the analysis of the PM_{10} fraction from road dust showed that the major component of coarse PM was mineral dust and unspeciated PM accounting for 55% followed by OC with about 21% and 1.05% of EC. Because it is still poorly constrained we did not consider any chemical speciation for road dust emissions in the model.

We also evaluated the modeled surface concentration of O_3, NO_x, and CO with AQMN data for all experiments. Modeled concentrations for primary species NO_x and CO, were similar for both experiments (Fig. 2 and Table 3). Both experiments underpredict SO_2, with a smaller bias in the MOZART experiment. This could be due to MOZART including more SO_2 formation pathways than RACM (Emmons et al., 2010; Stockwell et al., 1990). During both seasons the model is underestimating SO_2 and

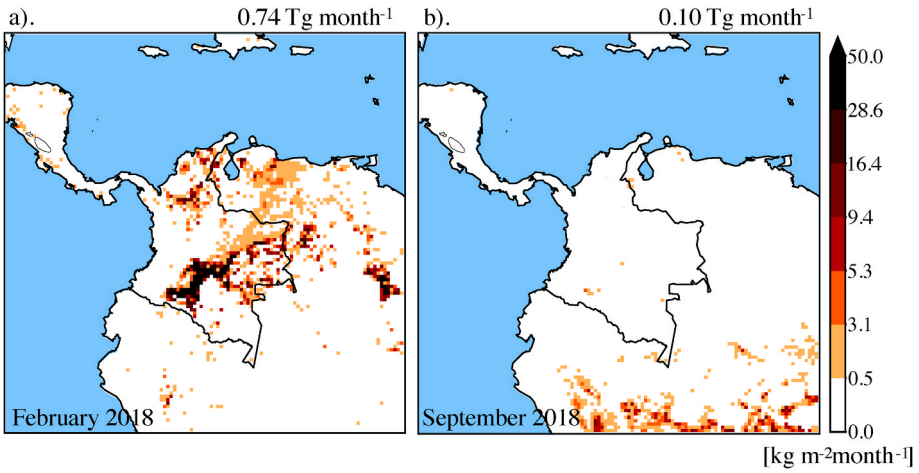


Fig. 6. FINN aerosol biomass burning emissions during (a.) February 2018 and (b.) September 2018.

Table 3
Monthly mean, mean bias (MB), normalized mean bias (NMB in %) and correlation coefficient (r) between observations at AQMN and modelling results over Bogotá.

Period	variables	Mean Obs.	RACM/MADE-VBS			MOZART/MOSAIC		
			MB	NMB	r	MB	NMB	r
February	PM2.5 [$\mu g\ m^{-3}$]	26.00	−2.73	−10.48	0.27	4.84	18.62	0.37
	O3 [ppb]	22.78	18.45	80.99	0.73	26.88	117.99	0.59
	NOx [ppb]	39.81	0.31	0.77	0.62	0.86	2.17	0.63
	SO2 [ppb]	2.19	−0.63	−28.65	0.35	−0.44	−20.19	0.38
	CO [ppm]	0.71	−0.06	−8.82	0.63	−0.05	−6.34	0.65
September	PM2.5 [$\mu g\ m^{-3}$]	16.64	−3.83	−23.00	0.57	0.02	0.12	0.53
	O3 [ppb]	21.23	6.31	29.72	0.29	9.10	42.87	0.41
	NOx [ppb]	42.31	−8.53	−20.16	0.58	−7.39	−17.46	0.53
	SO2 [ppb]	2.34	−0.97	−41.53	0.24	−0.85	−36.25	0.29
	CO [ppm]	0.84	−0.31	−37.26	0.53	−0.29	−34.96	0.54

CO with a normalized mean bias (NMB) of 24.4% and 7.57% for February, and 38.8% and 36.0% for September, respectively. Modeled NO_x is 1.5% higher than observations during February, but it is underestimated by 18.8% during September (Fig. 2).

The bias in SW flux at the surface is likely leading to stronger photochemistry and a resulting overestimation of the surface ozone concentration (Zhang et al., 2012) (supplementary material S9). This also suggests that including aerosol-radiation interactions, which were not considered in our model configuration, could further contribute to improve model performance. Even though both mechanisms overpredict hourly mean ozone concentrations, O_3 determined by RACM is 8 ppb lower than by MOZART (Table 3). Oxidation of NO with Hydroperoxyl radical ($NO + HO_2 \rightarrow NO_2 + HO$) can account for 70% of O_3 production (Jacob, 1999). This reaction has a higher rate constant in MOZART (Emmons et al., 2010; Stockwell et al., 1990), explaining the discrepancy between the two mechanisms. As in many other regions, the O_3 concentrations are lower within the urban perimeter compared to its surroundings due to the high VOCs and NO_x emissions, which causes O_3 to be titrated (Pirovano et al., 2012).

5. Conclusions

We examined chemical transport model simulations and detailed in-situ fine aerosol chemical composition data during low and high concentration seasons of 2018 over Bogotá Colombia. The modeling and observational data were used to understand the contribution of OA and its main source regions.

The field campaign data shows a persistent organic fraction in the $PM_{2.5}$ samples analyzed. This is one of the first published $PM_{2.5}$ chemical composition data for the city of Bogotá. On average, OA constituted 51% of $PM_{2.5}$ during February 2018 and 49% during September 2018. Elemental carbon concentrations were 7% of the total $PM_{2.5}$ at the two suburban sites, but were 23% of the $PM_{2.5}$ at the sampling site located within the city limits, showing a significant contribution from mobile sources. The severe underestimation of organic and elemental carbon occurred at this urban site suggests missing emissions in the inventory or an underestimation of both species in the speciation profiles. The fraction of unspeciated $PM_{2.5}$, likely corresponding to mineral dust from road dust resuspension, was also found to be important contributing to 23 and 28% of the total $PM_{2.5}$ in February and September, respectively.

The field data, together with the regional-scale modeling in this study demonstrates that even during the low biomass burning activity period of July–September, northern South America can still be heavily impacted by long-range transported plumes from northern Amazonia and other distant source regions.

The sensitivity of the modeled organic aerosol was analyzed by configuring the WRF-Chem model with two different aerosol mechanisms. Simulation results were evaluated against air quality monitoring site observations in Bogotá and chemical speciation from field campaigns. The simulated $PM_{2.5}$ correctly shows a strong seasonality, with a 10 μgm^{-3} difference between February and September. Consistently throughout both periods, the MOZART/MOSAIC experiment simulates 30% more fine particulate matter than RACM/MADE-VBS. Both mechanisms underestimate organic aerosol concentration at all sampling sites, but the simulations are consistent with the observations in that the organic fraction is dominant in the fine particulate matter in the city. Our results suggest the need to improve the chemical speciation of local emissions, including the chemical speciation of road dust to be implemented in future modeling analyses.

CRediT authorship contribution statement

Karen Ballesteros-González: Conceptualization, Writing – original draft, Investigation, Visualization. **Sebastian Orlando Espitia-Cano:** Investigation. **Maria Alejandra Rincón-Caro:** Investigation. **Juan Manuel Rincón-Riveros:** Investigation. **Maria Paula Perez-Peña:** Investigation. **Amy Sullivan:** Investigation, Methodology, Writing – review & editing. **Ricardo Morales Betancourt:** Conceptualization, Methodology, Writing – review & editing, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.atmosenv.2022.119161.

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